

Joonsuk Bae

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Experimental high-energy nuclear physicist building methodological infrastructure for precision hard-probe measurements. Main analyzer of the first ALICE Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (preliminary; paper in preparation; EPS-HEP 2025 jet working group merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics, Sungkyunkwan University 2022.03 – expected 2027.02
Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at $\sqrt{s} = 13.6$ TeV with ALICE*.
Defense expected 2026.12; degree conferral 2027.02.

B.S., Physics, Sungkyunkwan University 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN 2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory 2024.02 – present
LAMPS Collaboration, RAON 2022.03 – 2024.04

Research Highlights

- **Main analyzer**, first ALICE Run 3 inclusive charged-particle jet cross section at $\sqrt{s} = 13.6$ TeV: responsible for the full chain (O²Physics task, QA, unfolding, and a systematic-uncertainty program carried down to the track level); ALICE Preliminary; presented on behalf of ALICE at EPS-HEP 2025 (jet working group merge talk).
- **Tracking-efficiency systematics for Hard Probes 2026** (ALICE jet working group): derived the pp charged-particle tracking-efficiency uncertainty at $\sqrt{s} = 13.6$ and 5.36 TeV (p_T -binned, 2022/2023/2024 periods), combining the track-selection and ITS-TPC matching components.
- **ALICE O²Physics contributions** (AliceO2Group/O2Physics, 2023 – present): implemented the particle-level cross-section normalization, now becoming the standard normalization for Run 3 charged-jet analyses in the group, and corrected the shared track momentum-resolution smearing implementation.
- **ePIC Barrel Imaging Calorimeter**: lead analyst for the CERN PS T10 (2025.07) Pb/scintillating-fiber campaign; DAQ and prompt analysis at KEK AF-AR (2025.03); co-author on the 2024.08 campaign paper (submitted, NIMA-D-26-00482; arXiv:2604.22647), with module QA and assembly.

Selected Publications

INSPIRE: inspirehep.net/authors/2117232; ALICE Collaboration co-author since 2022.10; ECCE/ePIC and LAMPS collaboration papers are included in the INSPIRE record.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (submission expected 2026).
[main analyzer]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the BIC at the EIC,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; arXiv:2604.22647).
[co-author; module support and PMT/module performance characterization]
- Y. J. Kim et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682.
[co-author; Time-of-Flight maintenance and on-site checks]

Invited and Contributed Talks (since 2022)

Invited (external workshops, seminars).

- **FKPPN Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **HIM 2025**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Invited collaboration-meeting talks.

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09

- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed talks and posters.

- **Hard Probes 2026**, Nashville [international, talk] — *Jet-like dihadron correlations in pO, OO, and Ne-Ne collisions with ALICE* — 2026.06
- **Hard Probes 2026**, Nashville [international, poster] — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions* — 2026.06
- **EPS-HEP 2025**, Marseille [on behalf of the ALICE Collaboration; jet working group merge talk] — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.07
- **INPC 2025**, Daejeon [international, talk] — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE* — 2025.05
- **Quark Matter 2025**, Frankfurt [international, poster] — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **Hard Probes 2024**, Nagasaki [international, poster] — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2024.09
- **ATHIC 2023**, Incheon [international, poster] — *Dijet studies at the LHC* — 2023.04
- **Korean Physical Society (KPS) Spring and Fall meetings** [domestic] — 9 contributed talks and posters across Spring and Fall meetings, 2022.10 – 2026.04 (every meeting since joining ALICE).

Detector and Test-beam Experience

- **ePIC Barrel Imaging Calorimeter** — three beam-test campaigns — **CERN PS T10** (2025.07; lead analyst), **KEK AF-AR** (2025.03; DAQ, equalization, prompt analysis), and **CERN PS T10** (2024.08; module performance characterization); ran the PMT-module QA for the prototype modules and contributed Pb/scintillating-fiber bundle assembly; JANA2 / EIC-recon studies for sampling-fraction and energy response.
- **ALICE ITS2 alignment monitoring** (CTF level) — service work providing period-by-period tracking-performance diagnostics from compressed time-frame data.
- **LAMPS Forward Tracker** (2022 – 2024) — MPPC/scintillator-plane forward-tracking concept design and Geant4 feasibility simulations.

Service and Mentoring

- **Paper Committee member**, ALICE jet working group — OO charged-particle jet measurement led by W. Ham (SKKU).
- Co-mentor of W. Ham (M.Sc.), main analyzer of the ALICE Run 3 OO charged-jet measurement — jet reconstruction, unfolding, normalization, systematic uncertainties, and model comparisons; co-author with W. Ham on the corresponding ALICE Analysis Note.
- Co-organizer and instructor of the **KoALICE O^2 Physics Tutorial** (SKKU, 2024.01.22–23; ~22 participants from the Korean ALICE community; co-organized with H. Lee).
- Developed and shared the ePIC Korea-BIC beam-test analysis framework across the Korean BIC institutes (SKKU, PNU, KNU).

Technical Skills

Programming. C++, Python, ROOT, Geant4, JANA2 / EIC-recon, FastJet, Bash, Git, $\LaTeX$.

Software contributions. ALICE O^2 Physics — C++ analysis tasks on the columnar AO2D data model; contributions from jet finding to shared framework tools.

Event generators. PYTHIA 8 (Monash, CR modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.

Analysis. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

Languages. English (professional working proficiency), Korean (native).

References

Available on request.